

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 1: General Performance, Atmosphere (ISA), Wind & Runway

Syllabus Source: CAE Oxford (pp. 1-44) | Pass Target: >= 90% AviationExam

1. ATMOSPHERIC PROPERTIES & ALTITUDE CALCULATIONS

Aircraft performance is governed by air density, which dictates aerodynamic lift and engine thrust output:

- Pressure Altitude (PA): Altitude above the standard 1013.25 hPa datum plane: Pressure Alt = Elevation + 30 ft x (1013.25 - QNH) (or 27 ft/hPa).
- Density Altitude (DA): Pressure altitude corrected for non-standard temperature: Density Alt = Pressure Alt + 120 ft x (OAT - ISA Temperature). High density altitude severely degrades lift, climb rate, and engine thrust!
- ISA Temperature Formula: ISA Temperature (°C) at altitude h (thousands of ft) = +15°C - 2°C x (h / 1,000 ft) (up to tropopause at 36,090 ft / -56.5°C).

2. EASA REGULATORY FACTORED WIND RULES

When taking off or landing, wind provides aerodynamic assistance or penalty. Under EASA regulations, safety margins are legally mandated:

- Headwind Credit (50% Rule): Operators are permitted to take credit for NO MORE THAN 50% of the reported headwind component! (Factored Headwind = 0.50 x Reported Headwind).
- Tailwind Penalty (150% Rule): Operators MUST PENALIZE performance by AT LEAST 150% of the reported tailwind component! (Factored Tailwind = 1.50 x Reported Tailwind).
- Maximum Certified Tailwind: Standard certified tailwind limit for commercial transport jets is 10 knots (or 15 knots if specifically certified).

3. RUNWAY CONTAMINATION & HYDROPLANING PHYSICS

Runway surface conditions drastically alter friction coefficients and wheel braking authority:

- Dry Runway: Free of visible moisture; maximum braking coefficient.
- Damp / Wet Runway: Moisture present, water depth <= 3 mm. Reduces friction coefficient; landing distance factored by x 1.15.
- Contaminated Runway: More than 25% of the required runway surface is covered with standing water, slush, or loose snow exceeding 3 mm depth (or compacted snow / ice).
- Dynamic Hydroplaning Formula: Speed at which tires completely ride on water film with zero tire-to-pavement friction: $V_p \text{ (kt)} = 9 \times \sqrt{P}$ [P in psi], or $V_p \text{ (kt)} = 5.7 \times \sqrt{P}$ [P in bar].

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Density Altitude & Hydroplaning

Problem 1 (Density Altitude): Field elevation 4,000 ft, QNH 1003 hPa, OAT +25 C. -

Step 1: Pressure Altitude PA = $4,000 + 30 \times (1013 - 1003) = 4,300$ ft. - Step 2: ISA

Temp at 4,000 ft = $15 - (2 \times 4) = +7$ C. - Step 3: ISA Deviation = $+25 - (+7) = +18$

C. - Step 4: Density Altitude DA = $4,300 + 120 \times 18 = 6,460$ ft. Problem 2

(Hydroplaning Speed): Tire pressure 100 psi. - Formula: $V_{\text{hydro}} = 9 \times \sqrt{100} = 9 \times$

$10 = 90$ knots. - Above 90 kt on standing water (>3 mm), tire completely loses braking

friction!

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Factored Wind & Dynamic Hydroplaning Speed**

SCENARIO (Standard AviationExam Drill): A twin-jet is preparing for landing on Runway

27 (magnetic heading 270°): - Tower reported wind: 330° at 20 knots - Main gear tire

pressure: 144 psi QUESTION 1: What is the FACTORED HEADWIND component for performance

calculations? QUESTION 2: At what ground speed will dynamic hydroplaning occur if

runway has standing water? SOLUTION PASO A PASO (FOR DUMMIES): Part 1: Calculate

Reported Wind Components: - Wind angle to runway = $330^\circ - 270^\circ = 60^\circ$ off runway

heading. - Reported Headwind = Wind Speed $\times \cos(60^\circ) = 20 \text{ kt} \times 0.50 = 10.0$ knots.

- Reported Crosswind = Wind Speed $\times \sin(60^\circ) = 20 \text{ kt} \times 0.866 = 17.3$ knots. Part 2:

Apply EASA 50% Headwind Safety Factor: - Factored Headwind = 50% of Reported

Headwind = $0.50 \times 10.0 \text{ kt} = 5.0$ knots! - (You are ONLY allowed to credit 5 knots of

headwind assistance in your landing distance calculation!). Part 3: Calculate Dynamic

Hydroplaning Speed (V_p): - Formula: $V_p = 9 \times \sqrt{\text{Tire Pressure in psi}}$ - Tire

Pressure = 144 psi. $\sqrt{144} = 12$. - $V_p = 9 \times 12 = 108$ knots! FINAL ANSWER:

Factored Headwind = 5.0 knots. Dynamic hydroplaning occurs above 108 knots!

[!] EXAM TRAP (AVIATIONEXAM) - Headwind vs Tailwind Factoring Trap

- Headwind: CREDIT ONLY 50% (Multiply by 0.50). Gives a pessimistic, safe distance. -

Tailwind: PENALIZE BY 150% (Multiply by 1.50). Gives an extra-long, safe stopping

distance. - Never mix these up! Multiplying tailwind by 0.50 would be a fatal safety error!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA regulations, what proportion of the reported HEADWIND component may be credited for take-

- [A]: 100%
- [B]: Not more than 50%
- [C]: 150%
- [D]: Zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. EASA CS-25 requires that no more than 50% of nominal headwind component be used in performance graphs, providing a safety buffer against wind lulls.

Q2: An aircraft's main tires are inflated to 100 psi. What is the approximate dynamic hydroplaning speed on a r

- [A]: 100 knots
- [B]: 90 knots

- [C]: 60 knots
- [D]: 140 knots

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Formula: $V_p = 9 \times \sqrt{P} = 9 \times \sqrt{100} = 9 \times 10 = 90$ knots. Above 90 kt, a water wedge completely lifts the tire off the tarmac.

Q3: How does an UPHILL runway slope affect take-off performance and landing performance?

- [A]: Improves take-off and degrades landing
- [B]: Increases take-off distance (gravity opposes acceleration) and DECREASES landing distance (gravity assists deceleration)
- [C]: Has zero effect
- [D]: Decreases both distances

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An uphill slope acts as a retardant: it lengthens take-off ground run because gravity opposes engine thrust, but shortens landing roll because gravity helps decelerate the aircraft.